

BIELSKO-BIALA, Poland

WMO: 126000

Lat: 49.8067N

Lon: 19.0023E

Elev: 399

StdP: 96.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.6	-17.0	0.9	-13.6	-14.2	1.1	-10.2	15.2	5.4	13.2	3.6	2.3	60	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.4	19.8	27.5	19.1	25.8	18.3	20.9	27.1	19.9	25.8	19.0	24.5	2.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.3	23.8	17.8	13.4	22.6	17.0	12.7	21.5	62.0	27.1	58.7	25.9	55.7	24.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.4	7.9	-16.8	32.2	3.8	1.6	-19.5	33.3	-21.7	34.2	-23.8	35.1	-26.6	36.3		
(5)			-17.1	23.2	3.7	1.4	-19.8	24.2	-22.0	25.1	-24.1	25.9	-26.8	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-1.3	0.1	3.4	9.1	13.4	16.8	18.7	18.4	13.9	9.5	5.1	0.3		
	DBStd	8.36	5.59	5.53	4.73	4.68	3.89	3.70	3.40	3.38	3.72	4.54	5.03	5.24		
	HDD10.0	1454	352	277	207	69	12	1	0	0	8	66	161	301		
	HDD18.3	3539	610	510	462	278	158	73	38	41	140	274	398	558		
	CDD10.0	1088	0	1	4	43	118	203	268	261	124	51	13	1		
	CDD18.3	131	0	0	0	1	6	26	48	43	6	1	0	0		
	CDH23.3	1041	0	0	0	9	49	209	390	341	40	2	0	0		
	CDH26.7	247	0	0	0	1	4	47	102	87	7	0	0	0		
(14) Wind	WSAvg	3.1	3.7	3.6	3.5	3.1	2.7	2.5	2.6	2.4	2.7	3.1	3.4	3.8		
(15) Precipitation	PrecAvg	930	51	48	59	63	108	116	130	90	92	63	61	47		
	PrecMax	1366	110	110	140	142	433	244	348	201	217	162	109	121		
	PrecMin	708	18	15	21	0	34	41	18	20	30	12	0	12		
	PrecStd	151	21	23	29	30	73	52	71	41	48	38	26	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	13.6	17.6	24.7	26.9	30.5	31.8	32.1	27.2	22.3	17.9	13.4		
		MCWB	7.7	7.6	9.9	14.6	16.8	20.3	20.1	20.2	17.3	15.4	12.0	8.8		
	2%	DB	9.7	11.3	14.9	21.5	24.7	28.0	29.5	28.9	24.0	20.2	15.8	11.1		
		MCWB	6.6	6.8	8.5	12.8	16.4	19.7	20.0	19.7	16.7	14.5	11.0	7.6		
	5%	DB	8.0	9.5	12.8	18.9	22.8	25.8	27.3	26.9	22.2	18.4	14.2	9.0		
		MCWB	5.3	5.9	7.7	11.7	15.7	18.5	19.2	19.2	16.1	13.6	10.0	6.0		
	10%	DB	6.2	7.5	10.6	16.6	20.7	23.7	25.4	25.0	20.3	16.5	12.4	7.4		
		MCWB	4.0	4.7	6.5	10.4	14.6	17.3	18.3	18.2	15.2	12.6	9.0	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	8.5	10.4	15.3	18.7	21.8	22.2	22.0	19.1	16.5	12.7	9.2		
		MCDB	10.9	12.1	15.9	22.5	24.4	28.0	28.5	27.6	24.6	20.7	16.6	12.6		
	2%	WB	6.8	7.3	9.2	13.6	17.3	20.4	20.8	20.6	17.7	15.1	11.4	7.9		
		MCDB	9.4	10.6	13.7	20.2	22.8	26.2	27.0	26.9	22.6	19.0	15.3	10.8		
	5%	WB	5.5	6.1	8.1	12.3	16.2	19.1	19.9	19.6	16.7	14.0	10.3	6.3		
		MCDB	7.9	9.1	12.0	17.8	21.6	24.4	25.8	25.7	20.9	17.7	13.8	8.6		
	10%	WB	4.1	4.8	6.9	11.0	15.1	18.0	19.0	18.7	15.7	12.8	9.2	5.1		
		MCDB	6.0	7.4	10.2	15.9	20.0	22.8	24.3	24.1	19.5	16.1	12.0	7.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.6	6.8	8.8	9.1	8.9	9.2	9.5	7.9	7.0	5.7	5.1		
		MCDBR	6.3	8.2	10.3	12.2	12.4	12.1	12.5	12.2	10.9	9.6	7.8	6.5		
	5% WB	MCWBR	4.5	5.3	6.0	6.2	6.3	5.8	5.4	5.4	5.5	5.5	4.8	4.4		
		MCWBR	6.0	7.5	9.2	11.0	11.3	10.8	11.0	11.0	9.5	9.1	7.1	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.330	0.373	0.425	0.422	0.424	0.437	0.429	0.398	0.365	0.336	0.303		
	taud		2.301	2.271	2.233	2.162	2.218	2.246	2.216	2.244	2.302	2.373	2.386	2.367		
	Ebn at Noon		728	799	824	817	838	837	818	805	790	752	688	681		
	Edh at Noon		70	92	114	137	135	132	134	125	106	83	65	58		
(44) All-Sky Solar Radiation	RadAvg		0.87	1.60	2.70	4.02	4.83	5.15	4.98	4.50	3.18	1.93	0.97	0.68		
	RadStd		0.15	0.26	0.43	0.49	0.62	0.54	0.60	0.41	0.48	0.28	0.14	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	+1.07	N/A	N/A	N/A	N/A	N/A	+42
(47) Regional (42 neighbors)		+0.54	N/A	N/A	+0.82	+0.40	+0.34	-127	-168	+75	+30

Nomenclature: See separate page